BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 461: Introduction to Robotics

Examination: **Duration: 35min**
Semester : Summer 26 **Full Marks: 15**

Name:	ID:	Section:

1 A garments factory in Gazipur has installed an automated fabric handling robot named **"TEXBOT-9"**. During its design, the engineering team recorded the requirement of three different components, but they did not write down which transmission element was finally used in each. The recorded requirements are: **7.5**

P	A cutting head must **travel along a straight 3-metre rail** over the cutting table. It is **driven by a rotating servo motor**, and the movement must be highly precise while carrying a heavy cutting unit.
Q	The elbow joint of the arm is driven by a motor that is deliberately **mounted 1 metre away at the base** to keep the arm light. The transmission must be **light-weight, quiet and fast**, and it must run without any **grease or oil**, because even a single drop of lubricant will stain the white fabric.
R	Power coming from a horizontal shaft must be delivered to a vertical spindle at **a 90° angle**. The spindle runs at high speed under heavy load, and the factory floor requires quiet operation. The team noted that this part was the **costliest and the hardest** one to manufacture.

A. **Identify** the most suitable transmission component or gear for P, Q and R, and justify each choice in one line. **[1.5]**

B. A junior engineer proposes to use a pair of spur gears for R. **Explain** why this proposal fails, and **mention** one cheaper gear that could still do the job of S. **[3]**

C. The same company now wants a similar arm in the dyeing section, which is hot, dusty and chemically harsh, and runs at low speed with high torque. **Would** you keep the same element you chose for Q? If you want to change, **which** transmission system would you choose? **Justify** your answer using two properties. **[3]**

2	You are given a patch of a grayscale image with padding that you want to run convolution on using a 3x3 kernel. 	0	0	0	0	0
---	---	---	---	---
0	4	1	3	0
0	1	0	2	0
0	0	4	1	0
0	0	0	0	0 | Input given to the network | 1	0	-1
---	---	---
2	0	-2
1	0	-1	 Filter / Kernel A. The convolution is performed with a stride of 2. **Determine** the size of the feature map using the standard formula. **Show** the calculation. **[2]** B. **Compute** the complete feature map for stride 2. **[4]** C. Now **apply** a 2 × 2 average pooling operation on the feature map obtained in (c) and **write** the final output. **[1.5]**	**7.5**

Solution 1 :
A. P - Rack and Pinion - Converts rotational motion to linear motion with high precision and load capacity,
Q - Synchronous Belt - Light-weight, quiet and fast to transmit energy over a gap of 1 meter with no lubricants
R - Spiral Bevel gear - Cone-shaped gear that transmits power at 90°. Hardest and costliest to produce.

B. Spur gears are parallel-shaft gears. The teeth are straight and parallel to the shaft, so they can only connect two parallel shafts and cannot turn power through 90°. They are also noisy and not suited to high speed under load.
The cheaper 90° alternative is the straight bevel gear.

C. No. The belt should be replaced by a roller chain.

(i) Environmental resistance: the metal chain tolerates heat, dust and chemicals, while a belt degrades with temperature change and chemical exposure.
(ii) Load and precision: the chain is inelastic and stronger, so it suits low-speed, high-torque work and transmits motion with minimal slippage, whereas the belt would stretch and slip.

Solution 2 :

A. $W_{out} = (W - F + 2P)/S + 1 = (3 - 3 + 2\times1)/2 + 1 = 2/2 + 1 = 2$

B.

-2	2
-8	8

C. 0